Assignment - 1

Neural Network Architecture

A digital alarm should activate when exactly one of two binary sensors is ON. A single perceptron receives x_1 and x_2 uses $z = w_1 x_1 + w_2 x_2 + b$, and applies (Activation function) $H(z) = 1$ for $z \geq 0$ and 0 otherwise.

1 Write all four inequalities on w_1, w_2, and b ~~can classify every input correctly~~. required for perfect classification.

Ans Four Inequalities for Perfect classification

The perceptron is defined as : $z = w_1 x_1 + w_2 x_2 + b$

Activation function : $H(z) = \begin{cases} 1, z \geq 0 \\ 0, z < 0 \end{cases}$

The truth table is :

x_1	x_2	Desired output y	condition
0	0	0	$b < 0$
0	1	1	$w_2 + b \geq 0$
1	0	1	$w_1 + b \geq 0$
1	1	0	$w_1 + w_2 + b < 0$

Therefore, the four required inequalities are:

1. $b < 0$
2. $w_2 + b \geq 0$
3. $w_1 + b \geq 0$
4. $w_1 + w_2 + b < 0$

Q.2 Prove using those inequalities, that no choice of w_1, w_2 and b can ~~eat~~ classify every input correctly.

Ans Proof That a Single Perceptron cannot classify every input correctly

From the second inequality: $w_2 + b \geq 0$

From the third inequality :- $w_1 + b \geq 0$

Adding both inequalities gives :- $w_1 + w_2 + 2b \geq 0$

Since $b < 0$, we obtain: $w_1 + w_2 + b \geq -b > 0$

Therefore : $w_1 + w_2 + b > 0$

However, the fourth inequality requires:

$$w_1 + w_2 + b < 0$$

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This is a contradiction. Hence, a single perceptron cannot correctly classify all four inputs because the XOR function is not linearly separable.

Q.3 Construct a network with two hidden step neurons and one output step neuron. State all weights and biases.

Ans We can implement XOR using two hidden neurons and one output neuron.

The network equations are:

$$h_1 = H(x_1 - x_2 - 0.5)$$
$$h_2 = H(-x_1 + x_2 - 0.5)$$
$$y = H(h_1 + h_2 - 0.5)$$

Weights and biases

Neuron	Weights	Bias
• Hidden neuron h_1	$w_1 = 1, w_2 = -1$	-0.5
• Hidden neuron h_2	$w_1 = -1, w_2 = 1$	-0.5
• Output neuron y	$w_{h_1} = 1, w_{h_2} = 1$	-0.5

The first hidden neuron detects when $x_1 = 1$ and $x_2 = 0$. The second detects when $x_1 = 0$ and $x_2 = 1$. The output neuron activates when either hidden neuron activates.

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Q.4 ~~The~~ Calculate the forward pass for all four samples and verify the truth table.

Ans Using the equations:

$$h_1 = H(x_1 - x_2 - 0.5)$$
$$h_2 = H(-x_1 + x_2 - 0.5)$$
$$y = H(h_1 + h_2 - 0.5)$$

Sample 1: $x_1 = 0, x_2 = 0$

$$h_1 = H(0 - 0 - 0.5) = H(-0.5) = 0$$
$$h_2 = H(0 + 0 - 0.5) = H(-0.5) = 0$$
$$y = H(0 + 0 - 0.5) = 0$$

Sample 2: $x_1 = 0, x_2 = 1$

$$h_1 = H(0 - 1 - 0.5) = 0$$
$$h_2 = H(0 + 1 - 0.5) = 1$$
$$y = H(0 + 1 - 0.5) = 1$$

Sample 3: $x_1 = 1, x_2 = 1$

$$h_1 = H(1 - 0 - 0.5) = 1$$
$$h_2 = H(-1 + 0 - 0.5) = 0$$
$$y = H(1 + 0 - 0.5) = 1$$

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Sample 4: $x_1 = 1, x_2 = 1$

$h_1 = H(1 - 1 - 0.5) = 0$
$h_2 = H(-1 + 1 - 0.5) = 0$
$y = H(0 + 0 - 0.5) = 0$

Final verification table

x_1	x_2	h_1	h_2	Predicted y	Desired y
0	0	0	0	0	0
0	1	0	1	1	1
1	0	1	0	1	1
1	1	0	0	0	0

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